

The Analysis of KMV-Merton Model in Forecasting Default Probability

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Abstract— The paper gives brief review on two models called as Merton model and KMV-Merton model. Merton model is known as the triggers to the development of many credit risk models. Of all the credit risk models developed, the KMV-Merton model is the most popular. KMV-Merton model is developed to provide probabilistic assessment of firm's likelihood to default. Its ability in forecasting default for firms is proven when most of studies done by researchers and practitioners portray positive results. However in certain circumstances, the model becomes unavailable due to its structural constraints. Therefore, analysis is done on the KMV-Merton model so that modified default probability formulae are obtained for certain circumstances where the model becomes unavailable. Analysis is done using the mathematical approach. Then, the paper did data testing observation to justify the analysis. It is found that the results from the data testing satisfied the analysis done on KMV-Merton model. Thus, it verified that the modified default probability is true. The main contribution of this paper is it able to fill the gap exists in the KMV-Merton model in forecasting default for firms.

Keywords- credit risk; Merton model; KMV-Merton model; default probability; analysis.

I. INTRODUCTION

After years of globalisation, the world is still shocked by the bankruptcy news of biggest corporations such as General Motors Corporation, Lehman Brothers Holdings, and Enron Corporation. The collapse of these big corporations due to their failure in repaying the debt obligations has proved that credit risk remains the largest sources to the firms' bankruptcy. This is why credit risk needs to be managed efficiently and one of the ways is through the credit risk modeling.

The literature on credit risk modeling started since in the ninetieth century, when [1] introduced the extended model of [2] to price the simplest form of corporate liabilities. The extended model that called as Merton model is then enhanced by KMV for firms' probabilistic assessment of likelihood to default [3]. Since the model developed by KMV is based on the approach given by [1], thus the model is known as KMV-Merton model.

After the work of KMV, KMV-Merton model is used widely by practitioners, researchers and academicians in default studies. For example see reference [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], and [14]. Most of the studies

described the KMV-Merton model as a valuable default forecaster.

Unfortunately in certain circumstances, the model failed to forecast default. This is due to its structural constraints that limited the estimation of probability to default for firm that has zero value in the asset, debt or asset volatility. Accordingly, the purposes of this paper are:

1. to analyse and modify the KMV-Merton model for the cases where firms have zero value in the asset, debt or asset volatility; and
2. to justify the analysis and modification done on the model based on the data testing observation.

The rest of this paper is arranged as follows; Section II gives a brief introduction on the mathematical structure of KMV-Merton model. Section III explains the analysis and modification done on the model. Section IV presents the results and discussions. Section V concludes.

II. KMV-MERTON MODEL

In 1974, [1] introduced a formula to price firm's equity with significance information on default. [1] assumes that the firm's equity, E is a function of asset value, V that follows the lognormal random walk process under the Brownian motion. The equity behaves like a European call option with firm's debt, D as the strike price that defined as

$$E(V,0) = \max[0, V - D]. \quad (1)$$

The "max" is presenting the optionality of the equity's firm creditworthiness. The equity is supposed to follow the liquidation process, where the ownership equity paid only after all creditors are paid. If the firm fails to fulfill the promise payment, then the owner's equity would become zero.

The ability of the firm to make the repayment is depending on the value of asset at the expiry date, T . If the value of asset at time T is greater than the firm's debt, $V_T > D$, then the firm will pay the promised payment. Otherwise, the firm will not make the repayment since the value of equity is already worthless. The fact that bring by [1] about this failure has triggered the justification of firms' default.

In view of that, default is defined to occur when the market value of firm's asset is less than the book value of firm's liabilities by the time the debt matures [3]. Based on this definition, [3] derived a formula to estimate the default probability of firms describe as follows

$$P_t = N[-d]. \quad (2)$$

where P_t is the probability of default by time t and N is the cumulative probability distribution functions for a normal distribution d that is defined as

$$d = \frac{\ln\left(\frac{V}{X_t}\right) + \left(\mu - \frac{1}{2}\sigma_v^2\right)t}{\sigma_v\sqrt{t}} \quad (3)$$

The variable d is stand for distance to default. While V is the market value of firm's asset, X_t is the book value of firm's liabilities due at time t , μ is the expected return on firm's asset, and σ_v is the firm's asset volatility.

Based on (2) and (3) analysis and modifications are done through mathematical approach.

III. MODEL ANALYSIS AND MODIFICATIONS

The equations introduced in (2) and (3) are very useful formula used to forecast default probability of firms. However, there are certain circumstances where the formula is unable to be used due to its structural constraint.

As for equation (3), σ_v cannot be equal to zero since it can make the denominator of the equation becomes zero. Any numerator that is divided by zero is said to be undefined. As well as to the value of V and X_t , due to the attribute of the natural log function those variables cannot be equal to zero. Nevertheless, in reality these can happen. A firm may have zero asset volatility if its asset value is unchanged in certain period. It is possible the asset value may turn to zero due to the bankruptcy. The firm may also own zero liabilities due at time t .

Subsequently, analysis and modifications need to be done on (3) so that the KMV-Merton model becomes fully available in forecasting default. There is also the possibility when $V=0$, $X_t=0$ or $X_t=0$, $\sigma_v=0$. Yet this is typically rare. So analysis and modifications are done only on three basic cases discussed below:

A. When $V=0$

If the market value of firm's asset, V is found to be zero and other variables are greater than zero, thus the market value of firm's asset is said to be less than the book value of firm's liabilities, $V < X_t$. Supposed the firm is no longer having chances for recovery, thus the firm is said in bankruptcy.

In this condition, the firm is said to be fully default where the default probability is equal to 1 or 100%. This is because the total area of $N(-d)$ under the bell shape curve is equal to one.

To be specific, if $V < X_t$, then $\frac{V}{X_t} < 1$ and $\ln \frac{V}{X_t} < 0$. As $\ln \frac{V}{X_t}$ goes to $-\infty$, d would also go to $-\infty$. These can be expressed as follows:

$$\begin{aligned} & \lim_{\ln\left(\frac{V}{X_t}\right) \rightarrow -\infty} d \\ &= \lim_{\ln\left(\frac{V}{X_t}\right) \rightarrow -\infty} \frac{\ln\left(\frac{V}{X_t}\right)}{\sigma_v\sqrt{t}} + \lim_{\ln\left(\frac{V}{X_t}\right) \rightarrow -\infty} \frac{\left(\mu - \frac{1}{2}\sigma_v^2\right)t}{\sigma_v\sqrt{t}} \\ &= \frac{-\infty}{\sigma_v\sqrt{t}} + \frac{\left(\mu - \frac{1}{2}\sigma_v^2\right)t}{\sigma_v\sqrt{t}} \\ &= -\infty \end{aligned}$$

If d is approaching $-\infty$, thus P_t would approach to 1,

$$\begin{aligned} \lim_{d \rightarrow -\infty} P_t &= \lim_{d \rightarrow -\infty} N(-d) \\ &= \lim_{d \rightarrow -\infty} \frac{1}{\sqrt{2\pi}} \int_d^{\infty} e^{-1/2 d^2} dd \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-1/2 d^2} dd \\ &= 1 \end{aligned}$$

Therefore when $V=0$, the formula for P_t that satisfied the analysis is expressed as

$$P_t = N[-d] \quad (4)$$

with d defined as

$$d = \frac{\left(\mu - \frac{1}{2}\sigma_v^2\right)t - \ln X_t}{\sigma_v\sqrt{t}}$$

This is the modified default probability formula for a firm as $V=0$, while other variables are assumed to be greater than zero.

B. When $X_t=0$

If the book value of firm's liabilities due at time t is equal to zero, $X_t=0$ and other variables are greater than zero, thus the market value of firm's asset is said to be greater than the book value of firm's liabilities, $V > X_t$. When $V > X_t$, then

$\frac{V}{X_t} > 1$ and $\ln \frac{V}{X_t} > 0$. As $\ln \frac{V}{X_t}$ goes to $+\infty$, d would also goes to $+\infty$, that is shown as below

$$\begin{aligned}
& \lim_{\ln\left(\frac{V}{X_t}\right) \rightarrow +\infty} d \\
&= \lim_{\ln\left(\frac{V}{X_t}\right) \rightarrow +\infty} \frac{\ln\left(\frac{V}{X_t}\right)}{\sigma_v \sqrt{t}} + \lim_{\ln\left(\frac{V}{D}\right) \rightarrow +\infty} \frac{\left(\mu - \frac{1}{2}\sigma_v^2\right)t}{\sigma_v \sqrt{t}} \\
&= \frac{+\infty}{\sigma_v \sqrt{t}} + \frac{\left(\mu - \frac{1}{2}\sigma_v^2\right)t}{\sigma_v \sqrt{t}} \\
&= +\infty
\end{aligned}$$

Hence, the quantity P_t would approach to 0 as below

$$\begin{aligned}
\lim_{d \rightarrow +\infty} P_t &= \lim_{d \rightarrow +\infty} N(-d) \\
&= \lim_{d \rightarrow +\infty} \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{1}{2}d^2} dd \\
&= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{1}{2}d^2} dd \\
&= 0
\end{aligned}$$

This means the firm possesses the lowest probability to default that is extremely close to zero. Therefore, the formula for P_t that satisfied the analysis when $D = 0$ is as

$$P_t = N[-d] \quad (5)$$

with d defined as

$$d = \frac{\ln V + \left(\mu - \frac{1}{2}\sigma_v^2\right)t}{\sigma_v \sqrt{t}}$$

This is the modified default probability formula for a firm as $D = 0$, while other variables are assumed to be greater than zero.

C. When $\sigma_v = 0$

If firm's asset volatility is equal to zero, $\sigma_v = 0$ means the asset values are remained fixed for certain horizon where volatility is calculated. If this happens then no returns are generated at that time, $\mu = 0$. Thus, default would only depends on possibilities whether $V > X_t$, $V < X_t$, or $V = X_t$. In that way, the formula for P_t that satisfied is as below

$$P_t = N[-d] \quad (6)$$

with d defined as

$$d = \ln V - \ln X_t$$

This is the modified default probability formula for a firm as $\sigma_v = 0$, while other variables are assumed to be greater than zero.

IV. RESULT AND DISCUSSION

The paper considers a one year sample data from two Malaysian companies in 2008 obtained from DataStream, to verify the analysis and modified P_t formula stated in (4), (5) and (6). Debt is assumed to be due in one year, $t = 1$. The results are presented into five tables. Table I presents the one year default probability when $V = 0$, and Table II is when $X_t = 0$. While Table III, IV and V are for $\sigma_v = 0$ specifically when $V > X_t$, $V < X_t$, and $V = X_t$ respectively.

Table I and II apply the financial information from Company A. Whereas Table III, IV and V apply the financial information from Company B which is indeed has zero asset volatility. Only the value of asset and liabilities are switched in each table from the exact data, depending on the cases explained in Section III. The reason for this altering is because of the difficulty to find a company that follows the characters desired in each case. Furthermore, it is more reasonable to change the asset or liabilities rather than asset volatility. This is because asset volatility is calculated based on certain method and series of information that is not easy for us to predict.

TABLE I. The spreadsheet of one year default probability of Company A as $V = 0$

Item	Value
Asset value, V at $t = 31$ December 2008 (MYR mil)	0
Book value of liabilities, X_t at $t = 31$ December 2008 (MYR mil)	241.842
Asset volatility, σ_v	24.62%
Expected asset return, μ	3.45%
Distance to default, d	-22.276
Default probability, P_t by time $t = 31$ December 2008	100%

TABLE II. The spreadsheet of one year default probability of Company A as $X_t = 0$

Item	Value
Asset value, V at $t = 31$ December 2008 (MYR mil)	253.242
Book value of liabilities, X_t at $t = 31$ December 2008 (MYR mil)	0
Asset volatility, σ_v	24.62%
Expected asset return, μ	3.45%
Distance to default, d	22.497
Default probability, P_t by time $t = 31$ December 2008	2.2E-112

TABLE III. The spreadsheet of one year default probability of Company B as $\sigma_V = 0$ ($V > X_t$)

Item	Value
Asset value, V at $t = 31$ December 2008 (MYR mil)	100.493
Book value of liabilities, X_t at $t = 31$ December 2008 (MYR mil)	80.623
Asset volatility, σ_V	0.00%
Expected asset return, μ	0.00%
Distance to default, d	0.220
Default probability, P_t by time $t = 31$ December 2008	41.28%

TABLE IV. The spreadsheet of one year default probability of Company B as $\sigma_V = 0$ ($V < X_t$)

Item	Value
Asset value, V at $t = 31$ December 2008 (MYR mil)	80.623
Book value of liabilities, X_t at $t = 31$ December 2008 (MYR mil)	100.493
Asset volatility, σ_V	0.00%
Expected asset return, μ	0.00%
Distance to default, d	-0.220
Default probability, P_t by time $t = 31$ December 2008	58.72%

TABLE V. The spreadsheet of one year default probability of Company B as $\sigma_V = 0$ ($V = X_t$)

Item	Value
Asset value, V at $t = 31$ December 2008 (MYR mil)	80.623
Book value of liabilities, X_t at $t = 31$ December 2008 (MYR mil)	80.623
Asset volatility, σ_V	0.00%
Expected asset return, μ	0.00%
Distance to default, d	0.000
Default probability, P_t by time $t = 31$ December 2008	50%

Based on Table I, as $V = 0$ and others are non-zero values, d is found to be negative and P_t is equal to 100%. Contrast to Table II, which is as $X_t = 0$, d is found to be positive and P_t is equal to 2.2E-112, or in percentage it can be rounded up to 0%.

Although Table III, IV and V are constructed based on the zero asset volatility case, the values of d and P_t are found to be different from each other, except for μ that is equal to zero. Table III owns a positive d that is contradicted to negative d in Table IV. In the meantime, d is found to be higher in Table III compared to Table IV and V. Unlike P_t where the values are

found to be lower in Table III compared to Table IV and V. While the value of d and P_t in Table V is thought to be in the middle between Table III and IV, which are equal to 0 and 50% for d and P_t respectively.

As a whole, we found the increasing of d brings to the decreasing of P_t .

V. CONCLUSION

The KMV-Merton model has been widely used and studied by practitioners and researchers to forecast default for firms. It shows that in certain conditions, the model becomes unavailable due to its structural constraints. Therefore, analysis and modifications of KMV-Merton model are done in certain circumstances through mathematical approach. From the analysis, three modified default probability formulae are obtained. The formulae then were adapted to two Malaysian companies for data testing observation. Apparently, the data testing observation shows equivalent results with the analysis done on the KMV-Merton model. The results are summarized as follow.

Firstly, Table I and Table II show that when the market value of company's asset or the book value of company's liabilities is equal to zero, default probability becomes extremely low or high. This is because when $V_T = 0$, $V_T < D$ thus P_T is found to be high. Whereas when $D = 0$, $V_T > D$ thus P_T is found to be low. Briefly, when $V_T = 0$, $P_T = 100\%$ and when $D = 0$, $P_T = 0\%$.

Secondly, Table III, IV and V show that when the value of company's asset volatility is equal to zero, the expected asset return also becomes zero, while default probability can be categorized as high. In addition, default probability is said to vary according to the values of company's asset and liabilities in three conditions. In more specific, as the company's asset volatility is equal to zero, default probability found to be the lowest if $V > X_t$ and highest if $V_T \leq X_t$. While in the middle as $V = X_t$.

Lastly, all tables show that distance to default is inversely related to the default probability.

Literally, these justified our analysis done on the model. Thereby, it concluded that the modified default probability formulae are valid to be used in forecasting the default probability of companies that satisfied cases illustrated in this paper.

The analysis done is useful for academician, researchers or practitioners to understand the relationship of parameters used in KMV-Merton model to forecast default. As it is important to know better how the model works before upgrading the KMV-Merton model. The more essential is the formulae provided able to fill the gap exist in the KMV-Merton model in forecasting default for firms.

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REFERENCES

- [1] R. C. Merton, "On the pricing of corporate debt: The risk structure of interest rates". *Journal of Finance*, vol.29, pp.449-470, 1974.
- [2] F. Black, and M. Scholes, "The Pricing of Options and Corporate Liabilities". *Journal of Political Economy*, Vol. 81, No. 3, 637-654, 1973.
- [3] P.J. Crosbie, and J.R. Bohn, "Modeling Default Risk". Working paper, KMV LLC, 2003.
- [4] M. Vassalou, and Y. Xing, "Default Risk in Equity Returns" *The Journal of Finance*, Vol. LIX, No. 2, 831-868, 2004.
- [5] T.S. Bharat, and T. Shumway, "Forecasting Default with KMV-Merton Model". AFA 2006 Boston Meetings Paper. Available at SSRN:<http://ssrn.com/abstract=637342>, 2004.
- [6] J.L.M. Martin, and A.P. Trujillo, "Structural Models and Default Probability: Application to the Spanish Stock Market". *Journal of Investment Management and Financial Innovations*, vol.2, pp.18-26, 2005.
- [7] S.M. Lin, and J. Ansell, "Comparison of the Informational Bases Merton Type Model and Accounting Models to Assess Default within Small and Medium Sized Enterprises". Working Paper, Credit Research Centre, Management School and Economics, University of Edinburgh, 2006.
- [8] S.M. Lin, J. Ansel, and G. Andreeva, "Merton Models or Credit Scoring: Modelling Default of a Small Business". Working Paper, Credit Research Centre, Management School and Economics, University of Edinburgh, 2007.
- [9] C. Breig, and R. Elsas, "The Impact of Default Risk on Equity Returns: Evidence from a Bank-based Financial System". Working Paper, Institute for Finance and Banking, Ludwig Maximilians-University of Munchen, 2007.
- [10] L.J. Erken, "Case Study: KPN "The Merton model for default probabilities: a valid model for Dutch Companies?" Bachelor thesis finance, University of Van Tilbur, 2008.
- [11] Y. Lu, "Default forecasting in KMV". Master Dissertation, University of Oxford, 2008.
- [12] V. Fathi, "Credit Rating Methodology". Working Paper for Morningstar, 2009.
- [13] M.Si.M. Ardiansyah and A. Qoyum. "Default Risk in Islamic Equity Return (The Case of Kuala Lumpur Stock Exchange)". *Journal of Global Business and Economics*, vol 1, no 1, 2010.
- [14] S. Samsuddin, F.H. Tafri, A.H.M. Nawawi, N.A. Aziz, "Measuring the default risk of Sukuk holders for shariah compliance companies in Malaysia: Using Merton's model with maximum likelihood estimator," *Business*,